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Dry Eye Syndrome

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Introduction

Dry eyes, also known as dry eye syndrome (DES), dry eye disease (DED), ocular surface disease (OSD), dysfunctional tear syndrome (DTS), and keratoconjunctivitis sicca (KCS), are among the most common reasons for a visit to an eye doctor.[1][2] The definition of a dry eye according to the Tear Film and Ocular Surface Society (TFOS) Dry Eye Workshop II (DEWS II) is, "Dry eye is a multifactorial disease of the ocular surface characterized by a loss of homeostasis of the tear film, and accompanied by ocular symptoms, in which tear film instability and hyperosmolarity, ocular surface inflammation and damage, and neurosensory abnormalities play etiologic roles."[3]

The tear film is approximately 2 to 5.5 µm thick over the cornea and comprises 3 main components.[4][5] These components (lipid, aqueous, and mucin) are often described as layers, although this may oversimplify the tear film milieu.[6][7]

The tear film is approximately 2 to 5.5 µm thick over the cornea. It is composed of 3 main components, often described as layers (although this may be an oversimplification of the tear film milieu):[4][5]

1. **Lipid layer:** The most superficial layer; produced by the meibomian glands of the eyelids and functions to reduce the evaporation of tears [8]
2. **Aqueous layer:** The middle layer and the thickest component of the tear film; produced by the lacrimal glands, located in the orbits and the accessory lacrimal glands (glands of Krause and Wolfring) in the conjunctiva. Aqueous fluid contains water, metabolites, electrolytes, peptides, proteins, etc.[9]
3. **Basal layer:** Composed of mucins, or glycoproteins; predominantly produced by conjunctival goblet cells.[10] Mucins enhance the spread of the tear film over the corneal epithelium through the regulation of surface tension.[11]

Nomenclature and Terminology

DES and DED relate to a common ocular illness in which there is insufficient lubrication and moisture on the eye's surface. Numerous symptoms, such as dryness, irritation, burning, redness, and blurred vision, can be brought on by this illness. Even though the names are frequently used interchangeably, DED refers to a more comprehensive understanding of the disorder. It can be confusing because different medical practitioners and researchers use different terminologies.

DES

This term is frequently used informally to refer to inadequate tear production or poor tear quality, which causes ocular pain and associated symptoms. Patients and medical professionals frequently use it to characterize the disease.

DED

The multifaceted character of the ailment is better described by the more inclusive term DED. It recognizes that evaporative dry eye, aqueous deficit, and mixed etiologies are some underlying causes of dry eye, a complex and heterogeneous disease. DED covers a broader spectrum of clinical manifestations and causes.

Classification

DES is a broader term that emphasizes symptoms and a decline in tear quantity or quality rather than focusing on underlying causes in detail. DED acknowledges that the disorder has several facets and different subtypes. The 2017 TFOS DEWS II report offers a thorough categorization and in-depth explanation of dry eye illness.[3] Inflammation, neurosensory abnormalities, and environmental triggers are just a few of the contributing elements the report considers when classifying DED into aqueous-deficient and evaporative subtypes.

Diagnosis and Treatment

Diagnostic procedures and therapies for DES and DED are comparable. A thorough eye exam that includes tests for tear film quality and quantity, a review of the ocular surface, and an evaluation of the patient's stated symptoms is frequently required for diagnosis. Artificial tears, lubricating eye drops, dietary changes, antiinflammatory drugs, and, in extreme circumstances, procedures or operations may all be used as treatments. The underlying cause and severity of the condition determine the best course of action.

Research

Over time, dry eye research has advanced, and our understanding of the condition has become more complex. In contemporary medical literature and research investigations, the term DED is used more frequently to reflect this thorough understanding. Inflammation, meibomian gland dysfunction, and the effect of the environment on the ocular surface have all been studied in recent DED studies. Researchers hope to advance patient care by utilizing the term DED to cover various contributing factors.

Etiology

DED is usually classified into 2 major nonmutual exclusive etiology groups, which include evaporative dry eye and aqueous-deficiency dry eye.[12] Excessive evaporation from the ocular surface can occur due to a reduced muco-aqueous component in the tear film, whereas aqueous deficiency is due to reduced production, usually based on underlying autoimmune or systemic conditions. Mixed cases of the 2 types can also be found in patients with dry eyes.[13]

Numerous potential etiologies may contribute to the development of DED, and many cases may be multifactorial.[14] These include local ocular factors, systemic diseases, sociodemographic factors, environmental conditions, and iatrogenic causes such as medications or surgeries.[15] Potential causes or factors associated with dry eye include:

- **Systemic medications:**
  - Antihistamines
  - Antihypertensive
  - Anxiolytics/benzodiazepines
  - Diuretics
  - Systemic hormones
  - Nonsteroidal antiinflammatory drugs
  - Systemic or inhaled corticosteroids
  - Anticholinergic medications
  - Isotretinoin (causes meibomian gland atrophy)
  - Antidepressants [16]
- **Topical medications:** Including glaucoma drops or preservative toxicity from eye drops containing preservatives [17][18]
- **Skin diseases:** On or around the eyelids, such as rosacea or eczema [19]
- **Meibomian gland dysfunction:** A common comorbidity with thickening and erythema of the eyelids and inadequate or altered secretions of meibomian glands [20]
- **Ophthalmic surgery:**
  - Refractive surgery
  - Cataract surgery
  - Keratoplasty
  - Lid surgery [21]
- **Chemical or thermal burns:** Those that scar the conjunctiva [22]
- **Ocular allergies** [23]
- **Decreased androgen levels:** Ocuuring with menopause and other conditions [24]
- **Computer or device usage:** This may lead to decreased blinking when looking at the screen.[25]

- **Excess or insufficient dosages of vitamins:** Particularly vitamin A deficiency; can lead to xerophthalmia and the appearance of Bitot spots on the conjunctiva in severe cases [26]
- **Decreased sensation in the cornea:** Due to:
  - Long-term contact lens wear
  - Herpes virus infections
  - Other causes of a neurotrophic cornea [27]
- **Graft-versus-host disease** [28]
- **Autoimmune systemic diseases:**
  - Sjogren syndrome
  - Graves' ophthalmopathy
  - Connective tissue disorders such as:
    - Rheumatoid arthritis
    - Lupus
    - Thyroid disease [29]
- **Systemic conditions that increase the risk factors of dry eye:**
  - Diabetes [30]
  - Xerophthalmia [31]
  - Hereditary diseases [32]
  - Gut dysbiosis [33]
  - Multiple sclerosis [34]
  - Nerve damage pathologies [35]
- **Environmental factors:** Including exposure to irritants such as:
  - Chemical fumes
  - Cigarette smoke
  - Strong winds
  - High temperature
  - Pollution
  - High altitude
  - Low humidity [36][37]
- **Behavioral habits:**
  - Smoking [38]
  - Alcohol [39]
  - Poor sleep [40]
  - Unhealthy diet [41]
  - Dislipidemia [42]

Epidemiology

Dry eye is more common in women than men (due to female hormonal effects on the lacrimal and Meibomian glands and ocular surface) and has an increased prevalence with age.[43] Studies have shown that female gender is a risk factor in developing dry eye, with a prevalence that ranges from 12% to 22%.[15] The prevalence of DED worldwide varies depending on the diagnostic criteria employed, which ranges from approximately 5% to 50% in population-based studies.[13]

DES has been shown to be as high as 70% in visual terminal users.[43][44] In general, it is more common in Black individuals and Asian populations when compared with White individuals,[45] although geographic, climatic, and environmental variations may also be significant factors.[43][46] Evaporative dry eye is considered the most common subtype of DED. There may be discordance between dry eye signs and symptoms, with signs being more prevalent and variable than symptoms.[43]

Pathophysiology

Dry eye has traditionally been classified into 2 categories: aqueous deficient and evaporative.[3][47] These 2 categories, however, are not mutually exclusive, and numerous patients have a combination of these mechanisms of DED.

Aqueous Tear Deficiency

This category of dry eye is characterized by inadequate tear production. The predominant causes consist of Sjogren Syndrome (primary or secondary); diseases, inflammation, or dysfunction of the lacrimal gland; obstruction of the lacrimal gland; and systemic drugs (ie, decongestants, antihistamines, diuretics, beta-blockers, etc).[48]

Evaporative Dry Eye

This category of dry eye is characterized by increased tear film evaporation and a deficiency in the lipid portion of the tear film. In this case, the quantity of tears produced is normal; however, the quality of tears causes excessive evaporation. This alteration is most frequently caused by meibomian gland dysfunction.

Meibomian Glands

Meibomian glands line the eyelid margins and secrete oils that become the lipid layer of the tear film and reduce the evaporation of tears. Meibomian gland dysfunction may be caused by inadequate secretion due to atrophy, dropout of the glands, or obstruction of the gland orifices. Other major causes of increased tear evaporation include poor blinking (low rate, incomplete lid closure), lid aperture disorders, vitamin A deficiency, contact lens use, and environmental factors (low humidity, high airflow).

Tear Film

A hallmark of DED is hyperosmolarity of the tear film,[49] which may damage the ocular surface directly or indirectly by inciting inflammation, tear film instability, epitheliopathy, and ocular surface neurosensory abnormality.[50] The normal osmolarity of the tear film is usually less than 300 mOsm/L, which has been reported to be as high as 360 mOsm/L in patients with DED.[51]

Hyperosmolarity of the tear film leads to a cascade of signaling events that releases inflammatory mediators (ie, tumor necrosis factor, cytokines, alarmins, interleukin 1 and 6, etc.).

It leads to damage to the ocular surface, which may further decrease tear film stability, leading to self-perpetuation of the disease in a vicious cycle.[52][53] Apart from hyperosmolarity, other factors may initiate this pathologic cycle, including ocular surface inflammation caused by conditions such as allergic eye disease, topical preservative toxicity, or xerophthalmia.[54]

History and Physical

DED may lead to a number of symptoms, ranging from mild to severe[51]:

- Stinging, burning, or a feeling of pressure in the eyes
- A sandy, gritty, or foreign body sensation
- Epiphora, or tearing, is a symptom that is often counterintuitive. This is due to dryness leading to pain or irritation that results in intermittent excess tearing or epiphora.
- Pain is a broad term, and sharp and dull pain can be described as localized to some part of the eye, behind the eye, or even around the orbit.
- Redness is a common complaint and is often made worse by the rebound effect of vasoconstrictors found in many over-the-counter eye drops designed to reduce redness. Vasoconstrictors may decrease redness for the short term by constricting the vessels of the episclera but can have a rebound effect and increased redness after the drops wear off in a relatively short time period.
- Blurry vision, particularly intermittent blurry vision, is a common complaint and may also be described as glare or haloes around lights at night.
- Vision fluctuation and difficulties in reading
- A sensation of heavy eyelids or difficulty opening the eyes
- Excessive blinking
- Eyelid twitching
- Dryness is a common problem for contact lens wearers, and irritation may make contact lenses uncomfortable or even impossible to wear.
- Tired eyes (closing the eyes may provide relief to some individuals with dry eyes)
- Inability to cry in severe DED

Evaluation

There is no single gold standard sign or symptom for diagnosing DED. Evaluation of symptoms and signs of DED is recommended, as signs may be present without symptoms and vice-versa.

Symptoms

A verbal history allows the nonscripted elicitation of dry eye symptoms. In addition, many questionnaires have been developed to screen for symptoms of DED. Using a validated questionnaire allows accurate quantification of symptoms as a screening tool and monitoring for progression and response to treatments. Several questionnaires exist, such as the Ocular Surface Disease Index (OSDI),[55] Dry Eye Questionnaire (DEQ-5),[56] and Symptoms Analysis in Dry Eye (SANDE),[57] and others, which may help assess dry eye symptoms. Many questionnaires also include questions about subjective visual function or disturbances that may be attributable to dry eye.[58]

Tear Stability

Tear film breakup time

Tear film breakup time (TBUT) is the interval between a complete blink and the first break in the tear film. This is most often performed in the clinic using a slit lamp microscope after instilling sodium fluorescein stain to enhance the visibility of the tear film. A cutoff of fewer than 10 seconds for the appearance of a patch in the tear film is often considered consistent with DED. Alternatively, a noninvasive tear breakup time can be measured without fluorescein using instrumentation that evaluates the reflections of patterns or rings from the tear film or interferometry to assess for the appearance of discontinuity of the lipid layer after a blink. Studies have shown corneal hyperalgesia in eyes with a shorter TBUT.[59]

Tear Volume

Tear Meniscus Assessment

Assessment of the tear meniscus is performed at the slit lamp by judging the inferior tear film meniscus height. This technique is simple to perform but is subject to poor intervisit repeatability.[60] Instrumentation has been developed for more objective measurement of the tear film meniscus but is not currently widely available in most clinics.

Schirmer Test

A Schirmer paper strip is folded at the notch with the shorter end hooked over the lateral lid margin to avoid cornea irritation while the patient rests with closed eyes.[61] The Schirmer I test is performed without topical anesthetic to measure basic and reflex tearing with less than 5 to 10 mm (depending on cutoff used) of wetting after 5 minutes of diagnostic of aqueous deficiency. Alternatively, a topical anesthetic can be administered. Then, residual fluid blotted from the inferior fornix before performing testing to measure basic secretion with less than 5 to 10 mm of wetting is considered diagnostic for aqueous deficiency.[62]

Phenol Red Test

Like Schirmer testing, a cotton thread dyed with phenol red is hooked over the temporal eyelid into the sulcus for 15 seconds while the patient rests with closed eyes.[63] When wet, the thread turns red with cutoff values ranging from less than 10 to 20 mm used clinically.

Ocular Surface Assessment

Fluorescein Staining

Fluorescein staining allows the assessment of corneal damage. A minimal volume of fluorescein is instilled into the tear film with optimal viewing 1 to 3 minutes later. Greater than five spots of staining are considered positive results, with various grading scales, such as the Oxford grading scale, also used.[64][65]

Lissamine Green Staining

Lissamine allows the assessment of conjunctival and lid margin damage and, to a lesser extent, corneal damage. Greater than nine spots is a positive result.[64] Lid wiper epitheliopathy, or staining of the lid margin, can be performed with a positive result as 2 mm or more staining in length or greater than 25% in sagittal width.[66]

Conjunctival Redness

Conjunctival redness, or hyperemia, is not specific to DED as this may result from any stimulus that results in conjunctivitis, including infective, allergic, chemical, or mechanical etiologies. Grading is generally determined subjectively by slit lamp examination, although some devices with automated grading or digital photography can also be used.[67]

Tear Film Assays

Tear film Osmolarity

Elevated osmolarity and increased variability of osmolarity of the tears are characteristics of DED. Osmolarity values typically increase with disease severity. Various cutoff values have been reported, with 308 mOsm/L used as a threshold to diagnose mild-to-moderate disease, whereas 316 mOsm/L has been used as a cutoff for more severe disease.[58] Studies have shown that high osmolarity levels can lead to proinflammatory effects on the ocular surface, with the secretion of inflammatory cytokines and metalloproteinases that can cause chronic epithelium dysfunction and induce apoptosis.[68]

Matrix Metalloproteinases

These proteases are found in the tears of individuals with dry eyes. Matrix metalloproteinase-9 (MMP-9) levels can be tested using a point-of-care test.[69]

Eyelid Evaluation

Blepharitis

Evaluation of the eyelids is a crucial part of the evaluation to determine any factors contributing to DED. The evaluation includes assessment for anterior blepharitis and demodex blepharitis, which are frequent comorbidities of DED.[70]

Lid Wiper Epitheliopathy

The portion of conjunctiva along the lid margin that contacts the ocular surface to spread tears has been termed the "lid wiper."[71] Lid wiper epitheliopathy, or staining of the lid wiper with fluorescein or lissamine green, may be seen more commonly in individuals with DED, presumably due to increased friction between the lid and ocular surface.

Meibomian Gland Evaluation

The evaluation of the meibomian gland structure can be performed with meibography.[72] Although the outline of meibomian glands can be seen at the slit lamp or with a penlight by transilluminating the everted eyelid, enhanced visualization is obtained using infrared imaging systems to perform meibography. Inspection of the meibomian gland orifices along the eyelid margin can be performed to detect external obstructions of orifices. Meibomian gland function can be assessed by evaluating meibum quantity, quality, and expressibility.[58] Expressibility is assessed by applying digital pressure along the eyelid margin with a clear meibum easily expressed from the normal eyelid. Themeibum is turbid or viscous in meibomian gland dysfunction and is not easily expressed.

Eyelid Blink and Closure

Incomplete blinking and nocturnal lagophthalmos can result in DED. Blink assessment can be performed with or without a microscope or video recording equipment.[73] Lagophthalmos can be estimated by having the patient gently close their eyes and assessing for incomplete closure.

Evaluation for Systemic Disease

Numerous systemic diseases may cause DED, particularly primary Sjogren syndrome and secondary Sjogren syndrome caused by other autoimmune conditions such as rheumatoid arthritis, lupus, progressive systemic sclerosis, and dermatomyositis.[54] Other systemic abnormalities such as Parkinson's disease, androgen deficiency, thyroid disease, and diabetes have also been associated with DED.

Evaluation for systemic disease causing secondary dry eye may be warranted if an underlying condition is suspected. A review of systems is indicated to screen for underlying systemic diseases. Sjogren syndrome may also involve the salivary glands leading to dry mouth and predisposing to periodontal disease, and other mucous membranes may be affected, such as vaginal, gastric, and respiratory mucosae.[74] Laboratory testing for Sjogren syndrome (antibodies to Ro/SS-A or La/SS-B), rheumatoid factor, and antinuclear antibodies.[75] Referral to a rheumatologist may be indicated, and some cases of Sjogren syndrome may require a salivary gland biopsy by an oral surgeon.

Treatment / Management

Treatment of DES is performed in a step-wise approach that may vary depending on the severity of the disease.[13]

Initial approaches include:

- Education about the condition
- Modification of the environment
    - Eliminating direct high airflow or fans
    - Reducing screen time
    - Taking frequent screen breaks
    - Using a humidifier
- Identification and elimination of offending topical and systemic agents
- Topical ocular lubricants
- Lid hygiene (warm compresses and lid scrubs)
- Oral essential fatty acid supplements

Therapy aims to decrease signs and symptoms, restore ocular surface homeostasis, and enhance quality of life.[76]

The next step of treatment options includes

- Preservative-free ocular lubricants
- Reversible punctal occlusion (punctal plugs)
- Night-time ointment or moisture goggles
- Device-assisted heating or expression of the meibomian glands
- Intense pulsed light therapy
- Topical anti-inflammatory medications (corticosteroids, cyclosporine, lifitegrast)
- Oral antibiotics (macrolide or tetracycline).[51]

Further treatment options include:

- Serum eye drops
- Oral or topical secretagogues
- Therapeutic contact lenses
- Amniotic membrane grafting
- Surgical punctal occlusion
- Tarsorrhaphy

Differential Diagnosis

Many conditions may evoke symptoms similar to those caused by DED.[58] Some conditions may also be associated with or lead to DED, such as allergic conjunctivitis, cicatricial conjunctivitis, filamentary keratitis, and neurotrophic keratitis. Identifying the underlying primary condition in these cases is key to reducing the progression of the disease and worsening of dry eye.

Differential diagnosis for DED include:

- Conjunctivitis (allergic, viral, bacterial, parasitic/chlamydial)
- Anterior blepharitis
- Demodex blepharitis

- Cicatricial conjunctivitis (Stevens-Johnson Syndrome, mucous membrane pemphigoid)
- Bullous keratopathy
- Contact lens–related keratoconjunctivitis
- Eyelid malposition (entropion, ectropion) or abnormality (trichiasis) leading to ocular surface disease
- Keratitis (interstitial, filamentary, contact lens–related, neurotrophic)

Prognosis

There are minimal published data describing the natural history of treated and untreated DED.[43] DED is often considered chronic, with periods of exacerbation due to intermittent contributing factors. Postsurgical dry eye (such as following cataract surgery or refractive surgery) often improves with time, possibly related to the regeneration of corneal nerves or reduction of ocular inflammation.[77]

Complications

Complications from DED range from mild to severe. Mild-to-moderate DED causes symptoms detailed above, including ocular irritation or visual disturbances. More severe diseases can result in corneal complications, including infectious keratitis, ulceration, and scarring, which may cause subsequent loss of vision.[78][79] Although causation has not been established, several nonocular associations exist with DED, including depression, sleep and mood disorders, dyslipidemia, and migraine headaches.[80][81]

Deterrence and Patient Education

Patients should be educated regarding environmental or behavioral modifications that can be performed to reduce DED. For example, fans, air conditioners, or heating vents may worsen DED. Blinking awareness training or taking intermittent breaks may reduce the dry eye effects of staring at digital devices. Artificial tears should be promoted, while contact lens use can preferably be limited. Patients may also be educated regarding dietary factors influencing DED, including supplementation with essential fatty acids.

Enhancing Healthcare Team Outcomes

The primary care provider, ophthalmic nurse, and pharmacist should educate patients on the prevention and basic treatments of dry eyes by limiting screen time, blinking often, using artificial tears, and keeping the home environment cool and moist. Eye care providers should work with patients' primary care providers or rheumatologists to investigate possible underlying systemic diseases in cases where an underlying systemic disease is suspected.

In cases of primary or secondary Sjogren syndrome, treatment of the underlying disorder is often needed to treat the ocular manifestations adequately. Early identification and management of patients with dry eyes are imperative in reducing ocular symptoms and irreversible complications. Caring for patients with dry eyes necessitates a collaborative approach among healthcare professionals to ensure patient-centered care and improve overall outcomes. Health professionals involved in the care of these patients should possess the essential clinical skills and knowledge to diagnose and manage dry eye syndrome accurately.

A strategic approach involving evidence-based strategies to optimize treatment plans and minimize adverse effects is equally crucial. Each healthcare professional must know their responsibilities and contribute their unique expertise to patients' care plans, fostering a multidisciplinary approach. Effective interprofessional communication is paramount, allowing seamless information exchange and collaborative decision-making among the team members. Care coordination is pivotal in ensuring a patient's journey from diagnosis to treatment and follow-up is well-managed, minimizing errors and enhancing patient safety.

By embracing these principles of skill, strategy, ethics, responsibilities, interprofessional communication, and care coordination, healthcare professionals can deliver patient-centered care, ultimately improving patient outcomes and enhancing team performance in managing focal onset seizures. The interprofessional team approach with all clinicians, nursing staff, and pharmacists communicating openly and sharing information is optimal for addressing patients with dry eyes syndrome.

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